

INSTITUTE OF ACTUARIES OF INDIA

EXAMINATIONS

20th December 2018

Subject CT6 – Statistical Methods

Time allowed: Three Hours (15.00 – 18.00 Hours)

Total Marks: 100

INSTRUCTIONS TO THE CANDIDATES

1. *Please read the instructions inside the cover page of answer booklet and instructions to examinees sent along with hall ticket carefully and follow without exception.*
2. *Mark allocations are shown in brackets.*
3. *Attempt all questions, beginning your answer to each question on a separate sheet.*
4. *Please check if you have received complete Question Paper and no page is missing. If so, kindly get new set of Question Paper from the Invigilator.*

AT THE END OF THE EXAMINATION

Please return your answer book and this question paper to the supervisor separately. You are not allowed to carry the question paper in any form with you.

- Q. 1)** During festival of Diwali, SkyKitchen, a leading provider of food on flights wants to sell food on three airlines Flydigo, Superjet and Kristara on Mumbai to New Delhi route. With all 3 airlines Skykitchen has varying income structure depending on the demand and hence will impact the payoffs to them.

The airlines believe that the demand is equally likely to be High (H), Medium (M) and Low (L) and estimates Skykitchen's profits under each possible scenario to be:-

	H	M	L
Flydigo	10	14	4
Superjet	50	40	6
Kristara	20	15	8

- i) Define Saddle Point. (1)
 - ii) Determine the minimax solution to the problem. (1)
 - iii) Determine the decision which will maximize the maximum profit. (1)
 - iv) Determine the Bayes criterion solution to this problem. (2)
- [5]

- Q. 2)** The number of claims per auto insured follows a Poisson distribution with mean λ . The prior distribution for λ has the following probability density function:

$$f(\lambda) = [(500/\lambda)^{50} e^{-500/\lambda}] / \lambda \Gamma(50)$$

The Company observes the following claims experience:

	Year 1	Year 2
Number of claims	75	210
Number of autos insured	600	900

The company expects to insure 1100 autos in year 3. Determine the expected number of claims in year 3. [4]

- Q. 3)** The time series X_t is assumed to be stationary and to follow an ARMA (2,1) process defined by:

$$X_t = 1 + 1235 X_{t-1} - 36^1 X_{t-2} + Z_t - 17 Z_{t-1}$$

where Z_t are independent $N(0,1)$ random variables.

- i) Determine the roots of the characteristic polynomial, and explain how their values relate to the stationarity of the process. (2)
- ii)
 - a) Find the autocorrelation function for lags 0, 1 and 2. (4)
 - b) Derive the autocorrelation at lag k in the form

$$\rho_k = \frac{1}{2} + \frac{1}{2} \left(\frac{1}{2} \right)^k$$
 (4)
- iii) Determine the mean and variance of X_t . (2)
- iv) What is the advantage of using autocorrelation over auto-covariance? (1)

- v) If Y_t denotes a stationary time series defined at integer times and Z_t are independent variables, derive expression for:-

a) $\text{Cov}(Y_2, Z_3)$

(1)

b) $\text{Cov}(Y_3, Y_3)$

(1)

[15]

- Q. 4) i)** What are the criterias for a risk to be insurable? (3)

- ii) An Indian insurer classifies car insurance into three risk categories viz., Luxury Cars (category 1), Passengers Cars (category 2) and commercial cars (category 3). The number of claims is assumed to have a Poisson distribution with unknown parameter λ . Data on the number of claims in each category over the last 5 years is provided below.

Category	Year				
	1	2	3	4	5
1	15	12	21	27	32
2	17	18	12	26	40
3	29	33	37	47	53

Prior beliefs about λ are given by a gamma distribution with mean 40 and variance 20. Find Bayes Estimate of λ under quadratic loss.

(4)

- iii) In a country Actuarial you are a consulting actuary assessing the claim amounts (in Rs.mn) of four major general insurance companies A, B, C and D. Corresponding claim amounts and number of policies are shown in the tables below.

Company	Total claim amount			
	Year (j)			
	1	2	3	4
A	53	58	47	55
B	71	78	71	80
C	94	63	85	74
D	55	63	80	66

Company	Volume of business(P_{ij})			
	Year (j)			
	1	2	3	4
A	17	20	18	21
B	27	21	29	22
C	14	17	15	21
D	33	46	41	27

- a) Calculate expected premium per policy for all the insurers, using the assumptions underlying the Empirical Bayes Credibility Theory (EBCT) Model 2. (10)

- b) Calculate with reasons the credibility factors for all insurers using the assumptions as above with claims amounts being in £ and volume of business being double as mentioned above. Assume 1£ = Rs.89 (approx.) (2)

- c) State the condition under which the credibility estimate as calculated in **part (b)** above would be equal to credibility estimate as calculated using the assumptions underlying the EBCT Model 1. (1)

[20]

- Q. 5) i)** Define likelihood ratio and saturated model. (2)

- ii) Define Pearson and Deviance residuals and comment on the difference. (3)

- iii) Derive in which case the above two residuals will be same. (2)
[7]

Q. 6) A coin is chosen at random from a large collection of coins, the probability of tossing a head with that randomly chosen coin is λ , where λ has a PDF $f(\lambda) = \alpha\lambda^{\alpha-1}$ defined on the interval $0 < \lambda < 1$ (it is assumed that $\alpha > 0$). Suppose that a coin is chosen at random from the collection of coins. Express the following in terms of α .

- i) The coin is tossed twice. X denotes the number of heads that occur. Find the marginal probability function of $P(X=0)$, $P(X=1)$ and $P(X=2)$. (3)
- ii) Suppose that there is one head and one tail observed on the first two tosses of the coin. Find the posterior density of λ and hence probability that the next toss will be a head. (3)
[6]

Q. 7) i) The Reinsurance Regulations of a country allows taking benefit of Reinsurance purchased by the insurer in the calculation of the solvency capital that is required to run the business.

Following is the criteria used for calculating solvency capital for agriculture line of business of an insurer:-

Capital required is 1.5 times maximum of:-

Value 1 = 20% X Max (Net premium, 50% of Gross Premium); OR

Value 2 = 30% X Max (Net Claims, 50% of Gross Claims)

In the Financial Year 2016-17, an Insurer wrote Gross premium of Rs 900 Crores (1 Crore = 10 Million) in agriculture however didn't take any reinsurance for the same. Their loss ratio was 75% in that year.

However, in Financial Year 2017-18, the insurer took 70% quota share (70% ceded to reinsurer) for their agriculture premium and wrote Rs 1000 crores of gross premium. In this their gross loss ratio is 90%.

- a) Calculate Capital required by Insurer in Financial year 2016-17. (3)
b) Calculate Capital required by Insurer in Financial year 2017-18. (3)
- ii) Calculate the return on capital employed i.e. Net profit / Capital employed in following years for Insurer. You can assume expenses, other incomes, commission on Reinsurance to be Nil.
- a) Financial year 2016-17 (2)
b) Financial year 2017-18 (2)
- iii) What could have been possible reasons for Regulator to give restricted benefit of Reinsurance in calculation of solvency capital? (2)
- iv) In Financial year 2018-19, the insurer also took stop loss reinsurance to protect their net premium. Following are the various details of the arrangement. Calculate the maximum loss expected to be incurred by following:-
- a) Insurer (2)
b) Stop Loss Reinsurer (2)

Expected Gross Premium	Rs 1200 Crores
Retention of Insurer	20%
Stop Loss Structure	80% xs 120% i.e. Reinsurer that bear all losses on net premium from 120% till 200%
Reinsurance Premium rate charged by Stop Loss Reinsurer	15%
Assumptions to be made	1. Gross loss ratio will not exceed 200% 2. There are no other expenses, incomes

- v) What could have been the possible reasons for Insurer to take quota share Reinsurance in 2017/18 and Stop Loss Reinsurance in 2018/19? (2)
[18]

- Q. 8)** i) Explain the disadvantages of using truly random as opposed to pseudorandom numbers and list methods for generation of random variates. (3)
- ii) Give an advantage and a disadvantage of the Box-Muller algorithm relative to the Polar method. (2)
- iii) An Actuary is using a simulation technique to estimate the probability (p) that number of cars crossing a toll per hour exceeds a given count. The Actuary has carried out 1000 simulations and has produced an estimate that $p = 0.39$. The variance of the simulated value is 0.10.

Calculate the minimum number of simulations that the Actuary will have to perform in order to estimate p to within 0.01 with 95% confidence. (3)
[8]

- Q. 9)** i) What are the key limitations of Chain ladder method? (2)
- ii) The claims paid to date on a Health Insurance policy are as follows:-

Amounts in INR '000

Policy year	Development Year			
	0	1	2	3
2014	598	450	300	180
2015	685	510	344	
2016	735	540		
2017	705			

Other information

- Historical and future Inflation can be assumed as 6% per annum.
- The Ultimate loss ratio has been estimated at 80% and claims are assumed to be fully run off at the end of development period 3.
- Annual Premiums written in 2017 were INR 25,00,000.

Estimate the outstanding claims arising from policies written in 2017 using:-

- a) The inflation adjusted chain ladder method (4)
- b) The Bornhuetter – Ferguson Method – adjust for inflation (4)
[10]

Q. 10) The random variable Z has a binomial distribution such that

$$P(Z = z) = \binom{n}{z} p^z (1-p)^{n-z} \quad (z = 0, 1, 2, \dots, n)$$

Let $Y = \frac{Z}{n}$

i) Write down an expression for $P(Y = y)$, for $y = 0, 1/n, 2/n, \dots, 1$ (1)

ii) Express the distribution of Y as an exponential family and identify the natural parameter and the dispersion parameter. (2)

iii) Derive an expression for the variance function. (2)

iv) For a set of n independent observations of Y , derive an expression of the scaled deviance. (2)

[7]
